AZURE AI FOUNDRY LAB – 3.1

Title :

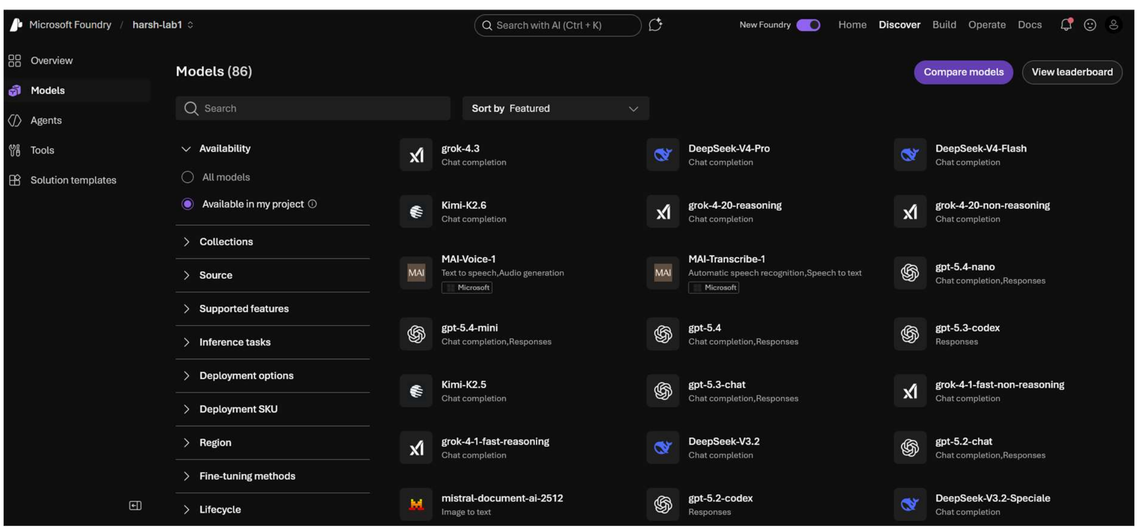
Choose and Deploy a Language Model

Objective :

The objective of this lab was to explore the Azure AI Foundry Model Catalog, compare multiple language models based on performance and cost metrics, analyze Quality Index and Accuracy scores, and identify the most suitable model for deployment based on project requirements.

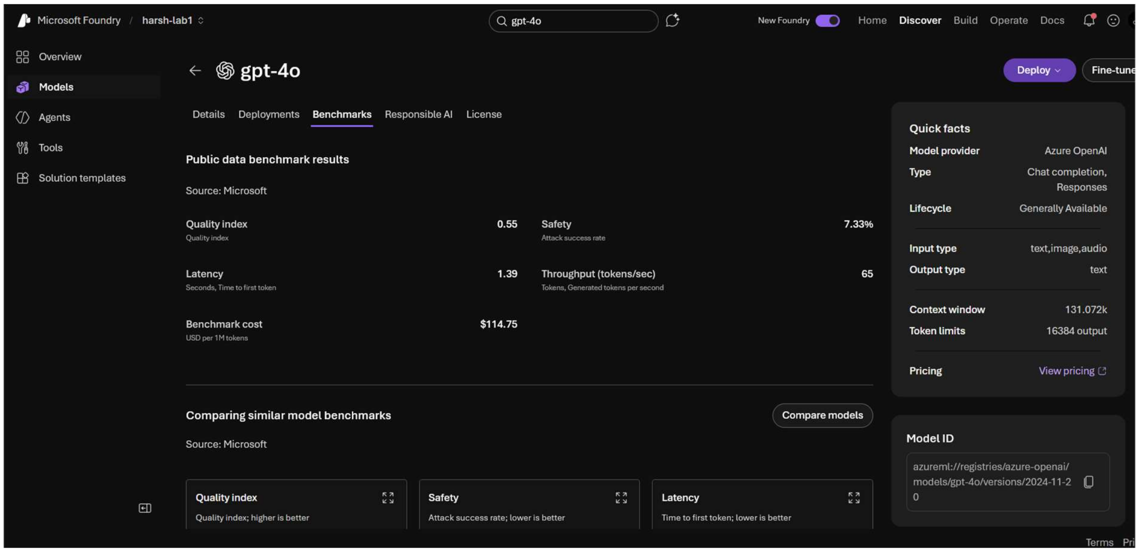
Tasks Performed :

1. Opened Model Catalog



Successfully accessed the Azure AI Foundry Model Catalog and explored the available language models for evaluation and comparison.

2. Explored Benchmark Metrics


Microsoft Foundry / harsh-lab1
gpt-4o
New Foundry
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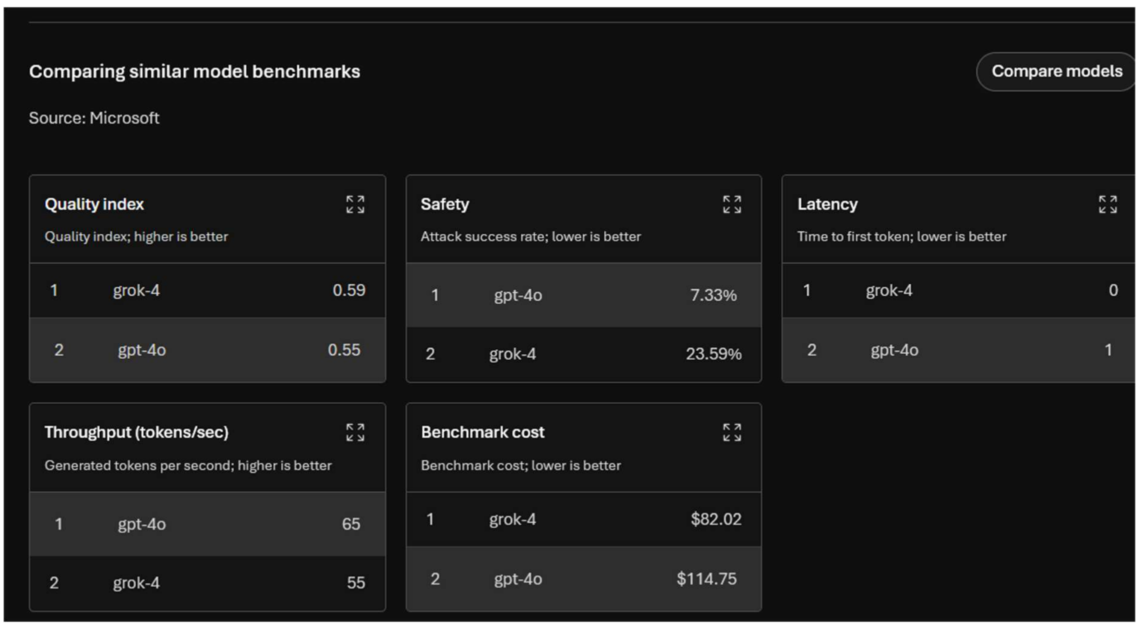
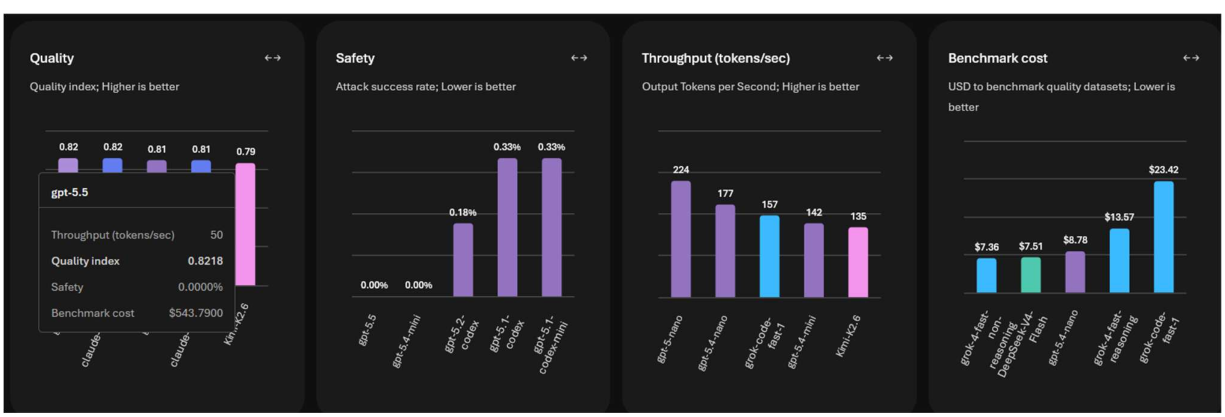
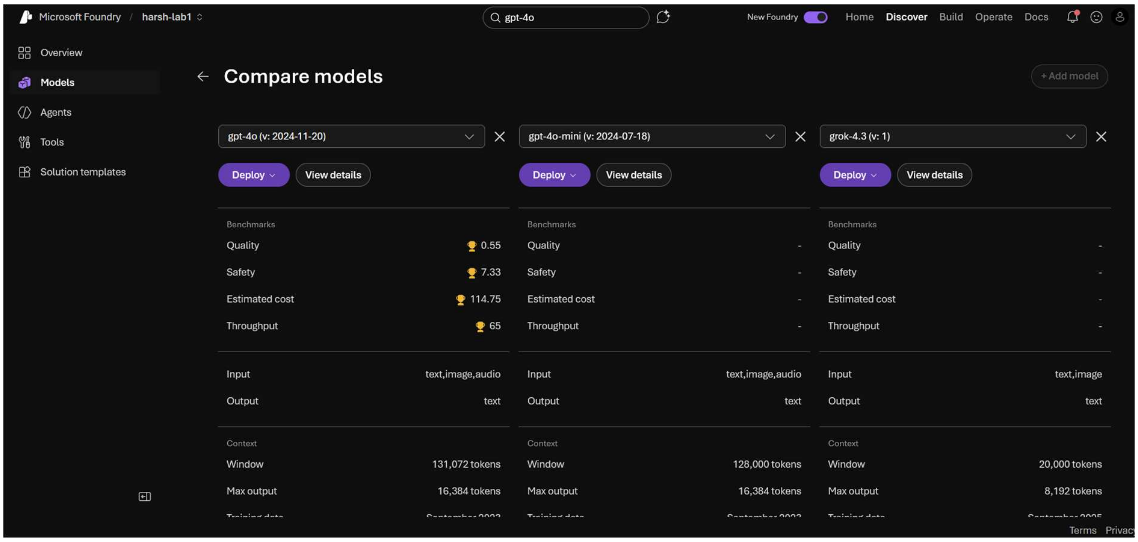
Tools
Solution templates
gpt-4o
Deploy
Fine-tune
Details
Deployments
Benchmarks
Responsible AI
License
Public data benchmark results
Source: Microsoft
Quality index 0.55
Quality index
Safety 7.33%
Attack success rate
Latency 1.39
Seconds, Time to first token
Throughput (tokens/sec) 65
Tokens, Generated tokens per second
Benchmark cost $114.75
USD per 1M tokens
Comparing similar model benchmarks
Compare models
Source: Microsoft
Quality index
Quality index; higher is better
Safety
Attack success rate; lower is better
Latency
Time to first token; lower is better
Quick facts
Model provider Azure OpenAI
Type Chat completion, Responses
Lifecycle Generally Available
Input type text,image,audio
Output type text
Context window 131.072k
Token limits 16384 output
Pricing View pricing
Model ID
azureml://registries/azure-openai/models/gpt-4o/versions/2024-11-20
Terms


Comparing similar model benchmarks
Compare models
Source: Microsoft
Quality index
Quality index; higher is better
1 grok-4 0.59
2 gpt-4o 0.55
Safety
Attack success rate; lower is better
1 gpt-4o 7.33%
2 grok-4 23.59%
Latency
Time to first token; lower is better
1 grok-4 0
2 gpt-4o 1
Throughput (tokens/sec)
Generated tokens per second; higher is better
1 gpt-4o 65
2 grok-4 55
Benchmark cost
Benchmark cost; lower is better
1 grok-4 $82.02
2 gpt-4o $114.75


Quality
Quality index; Higher is better
0.82
0.82
0.81
0.81
0.79
gpt-5.5
Throughput (tokens/sec) 50
Quality index 0.8218
Safety 0.0000%
Benchmark cost $543.7900
Safety
Attack success rate; Lower is better
0.00%
0.00%
0.18%
0.33%
0.33%
gpt-5.5
gpt-5.4-mini
gpt-5.2-codex
gpt-5.1-codex
gpt-5.1-codex-mini
Throughput (tokens/sec)
Output Tokens per Second; Higher is better
224
177
157
142
135
gpt-5-nano
gpt-5.4-nano
grok-code-fast-1
gpt-5.4-mini
Kimi-K2.6
Benchmark cost
USD to benchmark quality datasets; Lower is better
$7.36
$7.51
$8.78
$13.57
$23.42
grok-4-fast-non-reasoning
DeepSeek-V4-Flash
gpt-5.4-nano
grok-4-fast-reasoning
grok-code-fast-1

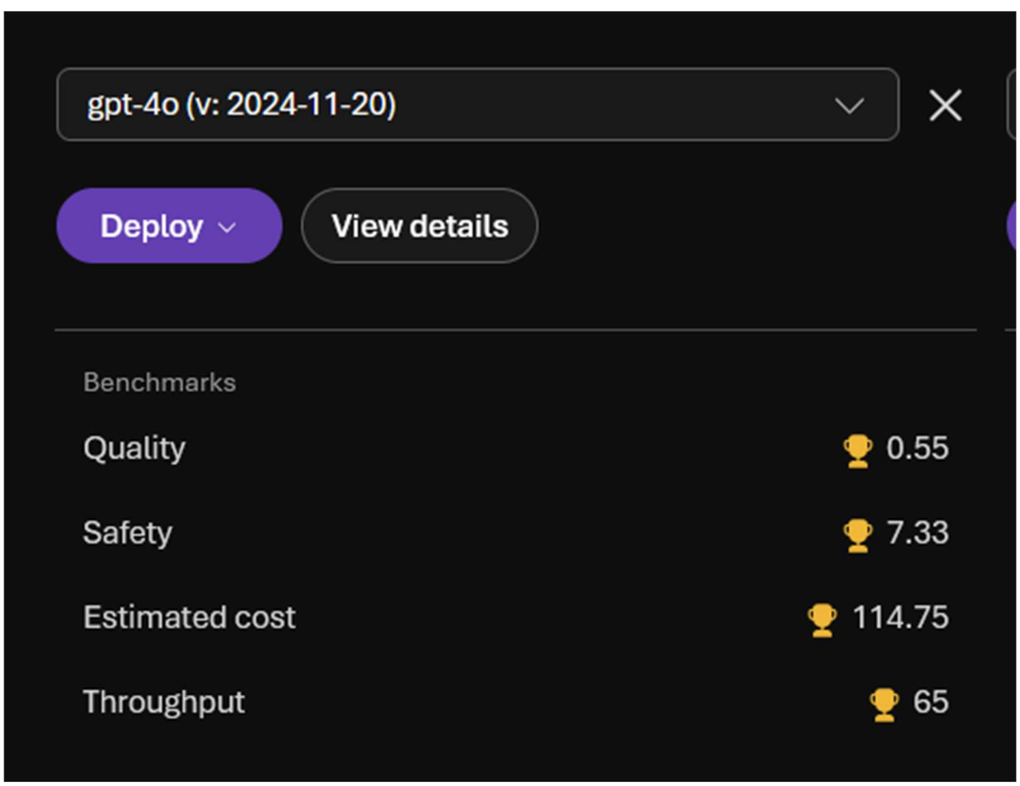
Compared GPT-4o and Grok-4 using Azure AI Foundry benchmark metrics including Quality Index, Benchmark Cost, Latency, Throughput, and Safety.

3. Compared Multiple Language Models



Compared GPT-4o, GPT-4o-mini, and another general-purpose language model using Azure AI Foundry's model comparison interface to evaluate quality, cost, safety, and throughput metrics.

4. Analyzed Benchmark Metrics :

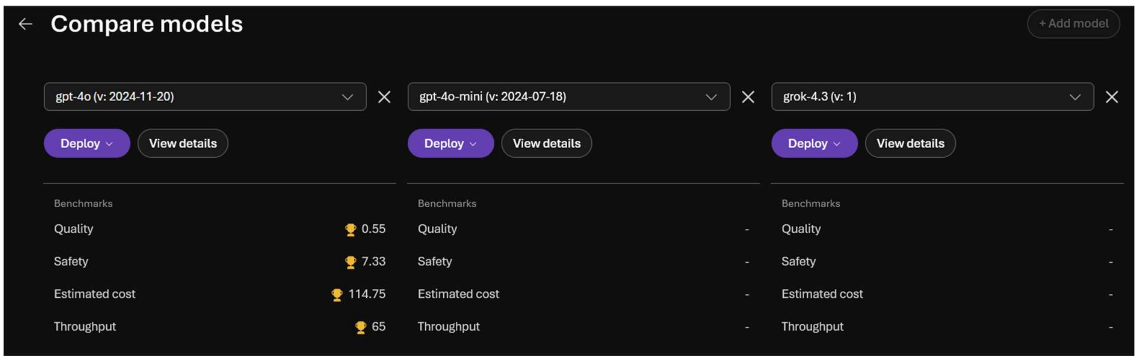


Observed benchmark metrics such as Quality Score, Safety Score, Estimated Cost, and Throughput to understand the strengths and trade-offs of each model.

5. Model Selection and Cost-Benefit Analysis

Three language models, GPT-4o, GPT-4o-mini and Grok-4.3 were compared using Azure AI Foundry.

Based on the benchmark comparison, GPT-4o was selected as the preferred model due to its strong performance, reliability, and suitability for general-purpose AI applications.



Observations :

• GPT-4o achieved a Quality Index score of 0.55.

• GPT-4o showed a throughput of 65 tokens per second.

• GPT-4o demonstrated strong benchmark performance across multiple evaluation metrics.

• GPT-4o Mini appeared to be a more lightweight and cost-efficient model.

• Comparing multiple models helped identify the trade-off between performance and deployment cost.

Result :

Successfully explored the Azure AI Foundry Model Catalog and compared multiple language models using benchmark metrics. The comparison included Quality Index, Safety, Estimated Cost, and Throughput. Based on the analysis, GPT-4o was identified as the most suitable model for general-purpose AI applications due to its strong overall performance and reliability.

Conclusion :

This lab provided practical experience in evaluating and comparing language models available in Azure AI Foundry. By analyzing benchmark metrics and comparing different models, it became easier to understand the trade-offs between performance, cost, and efficiency. The exercise demonstrated the importance of data-driven model selection when deploying AI solutions for real-world applications.

EXERCISES

Exercise 1 : Compare Models for Question Answering

Observation:
The Azure AI Foundry interface available during the lab provided benchmark-based model comparison and leaderboard rankings instead of the graph-based comparison view described in the exercise. Based on the benchmark metrics and model rankings, the following models were identified as strong candidates for question-answering tasks:

1. GPT-4o

2. GPT-4o Mini

3. Grok 4.3

These models demonstrated strong benchmark performance in terms of quality, throughput, and overall capability.

Exercise 2 : Cost-Benefit Analysis

Highest Quality Model:
GPT-4o

Lowest Cost Model:
GPT-4o Mini

Best Balance Between Cost and Quality:
GPT-4o Mini

Recommendation:
GPT-4o Mini offers the best balance between cost and quality for budget-conscious deployments, while GPT-4o remains the preferred choice when maximum performance and capability are required.